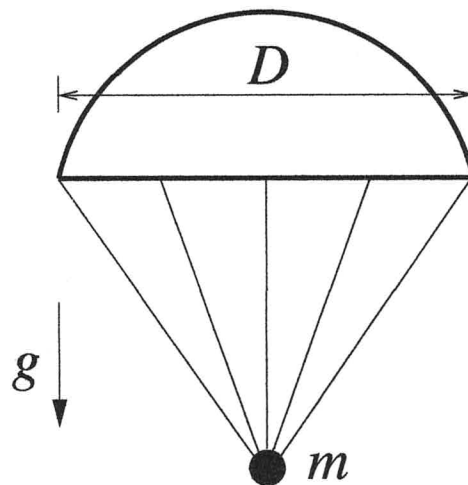


INTRODUCTION TO FLUID MECHANICS

FINAL (Jan 21, 2014)

Problem 3: We want to deposit a module of mass m over the surface of Mars ($\mu_M = 1.2 \times 10^{-5}$ kg/(m s), $\rho_M = 0.02$ kg/m³, and $g_M = 3.73$ m/s²) using a parachute of diameter $D_M = 5$ m. To assess the potential damage on landing, we need to know the terminal velocity of parachute descent over Mars V_M . For its determination, a series of experiments is carried out in a vertical wind tunnel on Earth ($\mu_E = 1.7 \times 10^{-5}$ kg/(m s), $\rho_E = 1.2$ kg/m³, and $g_E = 9.81$ m/s²), using the mission module attached to a model parachute, geometrically similar to that to be used in the Mars mission, but with a different diameter D_E . Use dimensional analysis to guide the design of the experiment and the analysis of the results, following the steps given below:

1. Use the Π theorem to express the dependence of V in terms of the minimum number of parameters.
2. Indicate the condition that needs to be satisfied for the effect of viscosity to be negligible. Adopt this simplification in the following.
3. Determine the diameter D_E needed to maintain physical similarity.
4. A terminal velocity $V_E = 2$ m/s is measured in the wind tunnel. Obtain the terminal velocity expected in the Mars mission V_M .
5. Verify that the viscosity is actually negligible for the experimental and mission conditions.



$$1) \quad V = V(\underline{\mu}, \underline{\rho}, \underline{g}, m, \underline{D}) \rightarrow \frac{V}{\sqrt{gD}} = f\left(\frac{\mu}{g^{1/2} D^{3/2}}, \frac{m}{\rho D^3}\right)$$

$$2) \quad \frac{\rho V D}{\mu} \gg 1 \quad 3) \quad \frac{m}{\rho_E D_E^3} = \frac{m}{\rho_M D_M^3} \rightarrow D_E = D_M \left(\frac{\rho_M}{\rho_E}\right)^{1/3} = 1.28 \text{ m}$$

$$4) \quad \frac{V_E}{\sqrt{g_E D_E}} = \frac{V_M}{\sqrt{g_M D_M}} \rightarrow V_M = \sqrt{\frac{g_M}{g_E}} \frac{D_M}{D_E} V_E = \left(\frac{g_M}{g_E}\right)^{1/2} \left(\frac{\rho_E}{\rho_M}\right)^{1/6} V_E \approx 2.44 \text{ m/s}$$

$$5) \quad \frac{\rho_E V_E D_E}{\mu_E} = 180.705 \gg 1, \quad \frac{\rho_M V_M D_M}{\mu_M} = 20333 \gg 1$$